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Stereological analysis of fractal fracture networks

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[1] We assess the stereological rules for fractal fracture networks, that is, networks whose fracture-to-fracture correlation is scale-dependent with a noninteger fractal dimension D. We first develop the general expression of the probability of intersection $p(l, l')$ between two populations of fractures of length l and l'. We then derive the stereological function that gives the fracture distribution seen in 2-D outcrops or 1-D scan lines for an original three-dimensional (3-D) distribution. The case of a fractal fracture network with a power law length distribution, whose exponent a is independent of D, is particularly developed, but the results can, however, be extended to any other length distributions. The analytical results were tested using a numerical model that generates 3-D discrete fractal fracture networks. The corresponding 1-D and 2-D length distributions are still described by a power law with exponents $a_{1\text{-D}}$ and $a_{2\text{-D}}$ that are related to the original 3-D exponent by $a_{3\text{-D}} = a_{1\text{-D}} + 2$ and $a_{3\text{-D}} = a_{2\text{-D}} + 1$, respectively, regardless of the fractal dimension. The density distributions of fractures in two or one dimensions remain fractal but with a dimension that depends on both the original 3-D distribution and the power law length exponent a. The fractal dimension of 2-D or 3-D fracture networks cannot be directly inferred from one-dimensional scan-line data sets unless a is known. We found a good adequacy between our predictions and measurements made on a few natural data sets. We propose also an original method for measuring the fractal dimension from the variations of the average number of fracture intersections per fracture. *INDEX TERMS:* 3250 Mathematical Geophysics: Fractals and multifractals; 3260 Mathematical Geophysics: Inverse theory; 5104 Physical Properties of Rocks: Fracture and flow; 8010 Structural Geology: Fractures and faults; *KEYWORDS:* fractures, fractal, scaling laws, stereological analysis, geometry

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1. Introduction

[2] In fractured rocks of very low matrix permeability the complex fracture pattern geometry is critical point to understanding the overall flow and transport properties. However, characterization of three-dimensional (3-D) fracture networks is severely limited by the virtual impossibility of locating, measuring, and analyzing reliable fracture data sets in situ. Consequently, most of the available data comes from either 1-D scan lines [*Du Bernard et al.*, 2002] or boreholes [*Barton and Zoback*, 1992] or 2-D outcrops [*Odling*, 1997; *Bour et al.*, 2002], whereas fracture systems should be characterized in three dimensions to provide a reliable basis for hydraulic properties modeling (Figure 1). The solution to that problem results in the assessment of the stereological rules that link 1-D or 2-D measurements with 3-D ones (Figure 2). Up to now, the stereological analysis has been mainly focused on fracture networks with a given length distribution and a uniform density distribution [*Marrett and Allmendinger*, 1991; *Piggott*, 1997; *Berkowitz and Adler*, 1998].

[3] However, numerous studies have demonstrated that fractures are spatially correlated, and that this correlation is well described by a fractal model [see *Bonnet et al.*, 2001, and references therein]. The role of the subsequent spatial clustering on the stereological rules has been poorly studied. *Malinverno* [1997] analyzed the possibility to estimate the volume distribution of turbidite beds from a power law size distribution of bed thickness sampled through well logs. *Borgos et al.* [2000] extended this work to the extrapolation of 1-D fracture length-frequencies distributions to higher dimensional samples. In particular, they analyzed the role of a nonuniform spatial distribution assuming that faults may be clustered in a self-similar way. For a power law length distribution of fractures, their analytical results suggest that the difference between the length distribution exponents, a, measured in two dimensions and in one dimension may be lower than 1, in contrast with previous work that assumed a uniform distribution of fracture density [*Marrett and Allmendinger*, 1991; *Piggott*, 1997; *Berkowitz and Adler*, 1998]. However, the rules for extrapolating a fractal dimension measured in d-dimensional space to higher dimensional space remains unknown.

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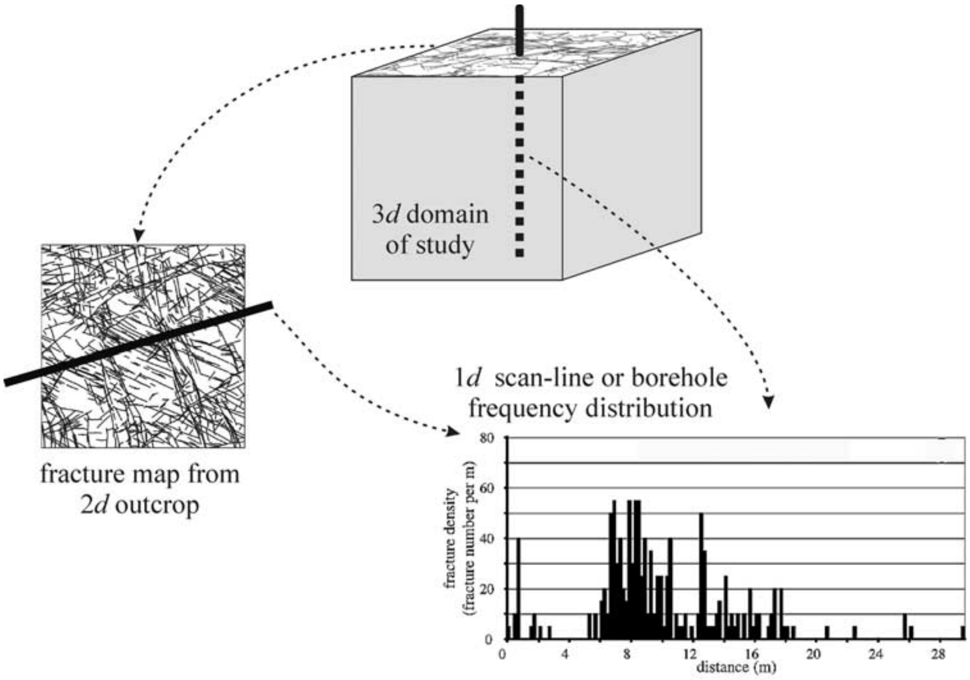


Figure 1. Schematic illustration of available data sets that basically define the 3-D fracture networks. (left) The 2-D fracture map is taken from an outcrop of the Hornelen Basin (Norway) [*Odling*, 1997]. (bottom right) The 1-D data are illustrated by a plot of the fracture density along a scan line. The example is taken from an outcrop across the Wadi Araba fault in the Suez rift [*Du Bernard et al.*, 2002].

[4] In the present work, we aim at characterizing analytically and numerically the stereological rules for stochastic fractal fracture networks. We study more especially 3-D fractal networks of circular disk shaped fractures whose length, here diameter, distribution is a power law for its relevance to natural fracture patterns [*Davy et al.*, 1990; *Bour et al.*, 2002]. However, our analytical results can be extrapolated to any other length distributions as long as the density is fractal. Our analytical results are checked with a numerical model that reproduces the fractal spatial correlations over a large range of scales. In the following, we first describe the statistical and numerical model of fractal fracture networks. We then focus on the analysis of the intersection probability between fractures in such correlated fracture networks. This approach provides the analytical developments required to tackle the stereological analysis. We finally discuss the consistency and consequences of our results to natural systems.

2. Fracture Network Modeling

2.1. Statistical Model of Fracture Network

[5] The main geometrical properties of fracture networks may be described by a simple statistical model, originally proposed by *Davy et al.* [1990], that is based on a double power law:

$$n(l, L) = \alpha L^D l^{-a} \qquad (1)$$

where $n(l, L)\mathrm{d}l$ is the number of fractures whose length is in the range $[l, l + \mathrm{d}l]$ and whose center belongs to a volume in three dimensions (or a surface in two dimensions) of typical

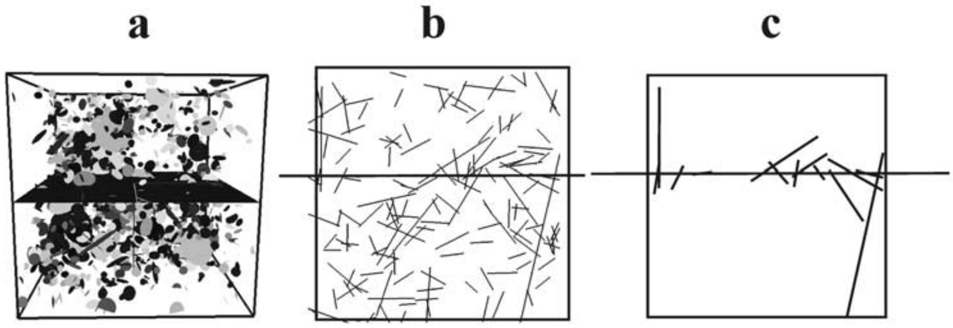


Figure 2. Schematic representation of the stereological analysis. (a) A 3-D network of disk-shaped fractures network, (b) a 2-D fracture traces network that may result from the intersection between the 3-D network and the 2-D plane, and (c) a 1-D fracture network intersecting a sampling scan line.

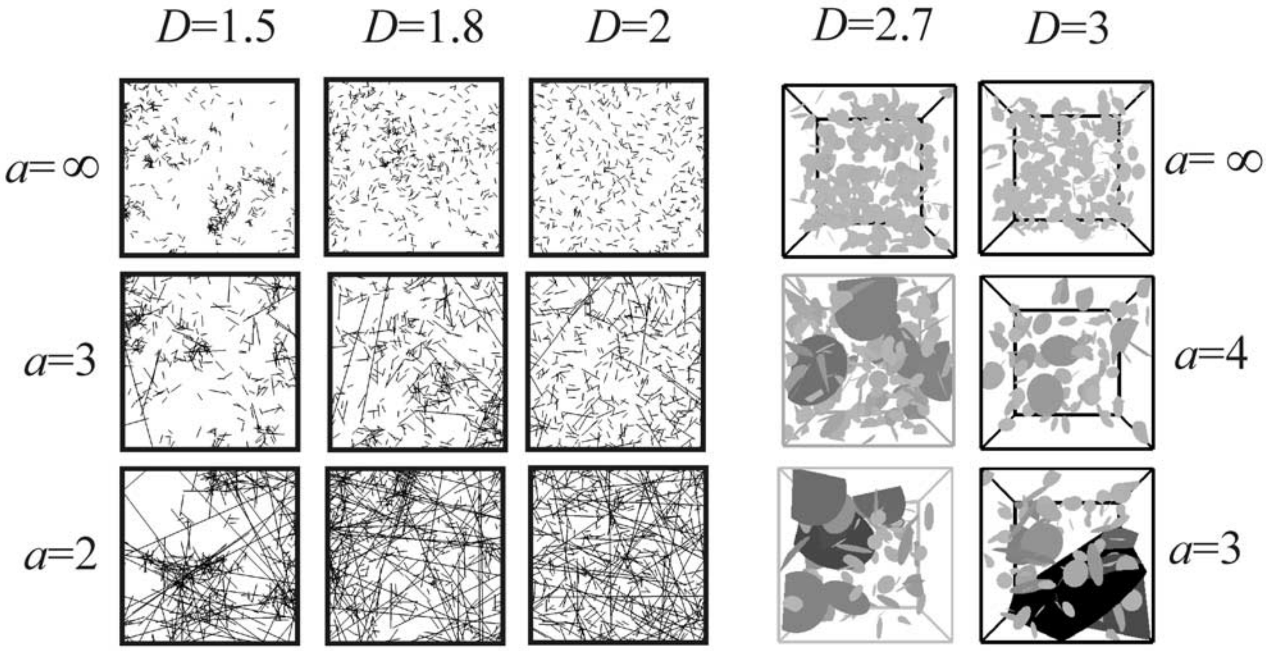


Figure 3. Examples of synthetic fracture networks generated with varying values of the exponent a (top to bottom) and of the fractal dimension D (left to right). The Euclidean dimension, d, is equal to 2 or 3. A decrease of a implies an increase of the proportion of large fractures. A decrease of D implies an increase of the clustering between fractures.

size L. The three parameters of the model are a the characteristic exponent of the fracture length distribution, D the fractal dimension that fixes the spatial distribution of fracture densities, and α a constant. The smaller is the exponent a, the greater is the proportion of large fractures with respect to the smaller ones. The fractal dimension D defines the scaling of the correlation between fractures, and thus of their clustering. Networks generated with different fractal dimensions are shown in Figure 3. In the model, D is equivalent to the correlation dimension [*Bour et al.*, 2002] (see a complete review by *Bonnet et al.* [2001]). The parameter α controls the density of fractures. The model has been validated on a natural 2-D fracture pattern, located in the Hornelen Basin (Norway), for which D = 1.8 and a = 2.75 describe the distribution from meter to kilometer scales [*Bour et al.*, 2002] (see also the outcrop shown in Figure 1).

[6] In three dimensions, we assume a population of disk-shaped fractures with diameters described by a power law distribution. The fracture traces are represented in two dimensions by straight-line segments. The model (1) does not describe the distribution of fracture orientations, but any specific orientation distribution can be added. In the following, we assume a random distribution of orientations. Equation (1) alone is a first-order model that says nothing about cross correlation between fracture parameters. Although correlations between fracture positions and fracture lengths have been reported in some cases [*Ackerman and Schlische*, 1997; *Bour and Davy*, 1999; *Darcel et al.*, 2003a], they are not considered in the present study.

[7] In the following we consider only those fractures whose centers lie within the volume or surface of typical size L. This assumption makes possible the calculation of very large statistically relevant systems. It also makes the analytical developments simpler. The complete development that includes long fractures whose center is outside the considered volume is derived in the appendix. Note that, because of the power law length distribution, this case is not rare and such finite size effects have to be treated explicitly.

[8] We consider in the following fracture networks having any fractal dimension D in the range [1, 3]. The range of variations of the length exponent a is kept to $[1, \infty]$ since an exponent a lower that 1 requires there be more large fractures than small ones in contrast to geological observations. When the exponent a tends toward infinity, all fractures have the same length.

2.2. Fracture Network Generation Method

[9] The numerical model developed generates networks of discrete fractures matching the statistical model described in equation (1). The method relies on a multiplicative cascade process that produces a probability distribution of fracture density with any fractal dimension in the range $[0, d]$, where d is the Euclidian dimension [see *Darcel et al.*, 2003b]. Each cascade corresponds to a subdivision of the total sampling domain in equally sized subdomains, and to a nonuniform probability distribution of fracture occupancy. The smaller subdomain has a volume arbitrarily fixed to 1, while the whole sampling domain has a volume up to $(1024)^3$ in three dimensions.

[10] The variety of fracture patterns that may be generated with the numerical model is illustrated in Figure 3. When a tends toward infinity, fractures are all of same length so that the clustering of fracture centers can be easily observed. When a is sufficiently small, the number of large fractures that may crosscut the system size increases and may partly mask the fractal correlation between fracture centers. These

geometrical features and their meaning are thoroughly discussed by *Darcel et al.* [2003b].

3. Stereology and Local Connectivity

[11] In practice, the stereological analysis consists in relating the geometrical properties of a real 3-D fracture network with the properties of its intersection with a 2-D outcrop plane, or with a 1-D scan line (Figure 2). The study is focused on the stereological rules that permit the extrapolation of the scaling exponents of equation (1), e.g., the length distribution exponent a and the fractal dimension D. The easiest way to solve the stereological problem is to derive in a forward sense the length distribution of the fractures that intersect a 2-D plane or a 1-D line given the length distribution of the entire fracture network. This can be performed analytically if the probability $p(l, L)$ for a fracture of length l to intersect a transect of size L, is defined. Indeed, the length distribution in a 2-D plane, or in a 1-D transect, is simply given by the product $n(l, L)\,.\,p(l, L)$ from which one should be able to derive the length distribution exponent a and the fractal dimension D.

[12] Analyzing the probability of intersection between a fracture (disk or segment) and a sample of observation (plane or scan line) of identical Euclidean dimension is equivalent to analyze the probability of intersection between fractures of respective length l and L. Thus the stereological issue is a particular case of the study of the connectivity between fractures of various lengths. For that reason, we first analyze the probability of intersection between fractures, the stereological analysis being studied in the next part of the paper. We first focus on the theoretical expression of the probability of intersection between fractures of the same length in a fractal network. We next derive the probability of intersection between fractures of different lengths. At last, we develop the stereological analysis, which becomes almost straightforward when previous relationships are stated.

4. Connectivity Between Fractures

4.1. Constant Length Fractures

[13] The probability of intersection between fractures uniformly distributed in space is given through excluded area arguments [*Balberg et al.*, 1984]. The excluded area (or excluded volume in three dimensions) between two objects is defined as the area (or volume) around an object into which the center of the other object cannot be if overlapping between the two objects is not allowed. Illustrations and development of this concept are given by *Balberg et al.* [1984] and *Adler and Thovert* [1999]. The normalization of the excluded area by the area of the system provides the probability of intersection between the two elements. Thus, in a 2-D system of size L, two distinct fractures of length l and l', have a probability of intersection given by [*Balberg et al.*, 1984]

$$p(l, l', L) = \frac{2}{\pi}\frac{l\,l'}{L^2}. \tag{2}$$

For identical fractures, equation (2) leads to $p(l, L) \sim (l/L)^2$, meaning that the probability of intersection between fractures is simply proportional to the square of the fracture length.

[14] In a fractal network, the probability of intersection should depend on the fractal dimension D due to spatial fracture clustering (Figure 3). The probability of intersection between two fractures can be determined analytically: it is given by the ratio between the average number of intersections of a fracture, $n_i(l, L)$, and the total number of fractures in the system (N_{tot}). In average, the number of intersections on a fracture is equal to the number of fractures that are included within its own excluded area. This area should not depend on the fractal distribution of fracture density but only on the shapes and orientations of the objects. Indeed, if a fracture center lies within the excluded area of another fracture, both will intersect whatever the fractal dimension of the system. However, because of fracture clustering, $N(l)$, the average number of fracture centers included within the excluded area (or equivalently within an area of diameter l centered on a fracture center) is given by

$$N(l) \propto N_{tot}\left(\frac{l}{L}\right)^D, \tag{3}$$

where N_{tot} is the total number of fractures included in the whole system of size L. Equation (3) corresponds simply to the definition of a fractal that states that the number of boxes of sizes l necessary to recover a fractal system of size L varies as $N_b(l) \sim (L/l)^D$ [*Mandelbrot*, 1982]. Assuming $n_i(l, L)$ to be proportional to $N(l)$, the probability of intersection between two fractures, $p(l, L) = n_i(l)/N_{tot}$, should be given by

$$p(l, L) \propto \left(\frac{l}{L}\right)^D. \tag{4}$$

Equation (4) is in accordance with the uniform case ($D = 2$), and with the fact that the intersection probability between fractures should increase when D decreases, due to fractal clustering (see the examples of synthetic fracture networks in Figure 3). The same reasoning applied in three dimensions leads to similar results with same expressions of equations (3) and (4).

[15] To check equation (4), we analyzed numerically the evolution of $p(l, L)$ with l and L over a range where $l \ll L$, so that finite size effects are negligible. Results of numerical simulations for D varying within the range [1.3, 3] are plotted on Figure 4a. As expected, for the same system size, the probability of intersection increases as D decreases, in perfect agreement with the increase of fracture clustering when D decreases (Figure 4a). To characterize the variations of $p(l, L)$ with L, we derived for each curve the scaling exponent, D' and reported it as a function of D (Figure 4b). Results show that $D' = D$, in complete agreement with equation (4).

4.2. Distinct Fracture Lengths

[16] Let's now consider the probability of intersection between fractures of distinct lengths, $p(l, l', L)$, in a fractal fracture network of dimension D. The analysis is first developed for 2-D systems, with l' and l the lengths of the largest and smallest fracture, respectively.

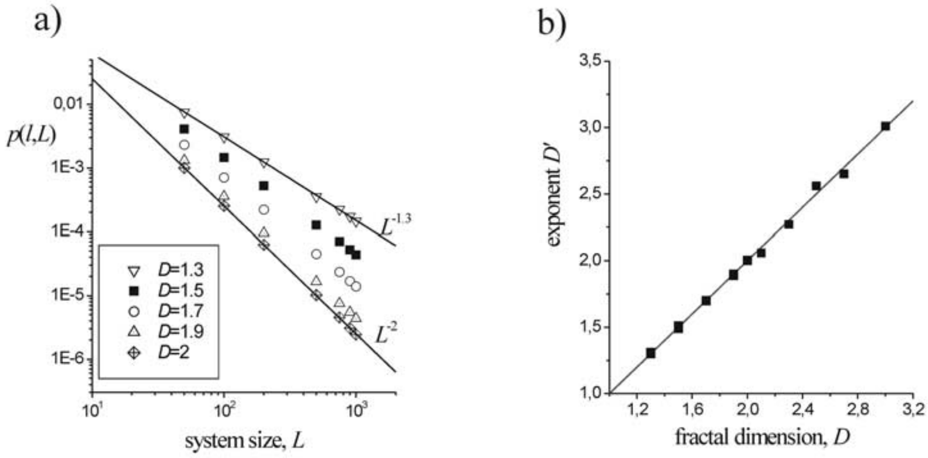


Figure 4. Probability of intersection between two fractures of length l, $p(l, L)$. (a) Evolution of $p(l, L)$ for several values of D. Solid lines correspond respectively to $L^{-1.3}$ and L^{-2}. The probability $p(l, L)$ varies as $p(l, L) \sim L^{-D'}$ with D' equal to the fractal dimension D. (b) Exponent D' as a function of D.

[17] The first remark which is useful to understand the following developments is that fracture length "randomizes" fractal correlation because a fracture length is itself a 1-D correlation which is not consistent with fractal correlations. In other words, if fracture centers are fractal as we assume in our model, all other points of the fracture cannot be fractal at least at scale smaller than fracture length. If both the large and the small fractures have a random orientation as it is assumed here, the "uncorrelated" domain, that is the one swapped by both orientation distributions, has an area (or a volume in three dimensions): $A_1 = \pi/4 * (l' + l)^2$ (Figure 5). Within this area, the connectivity does not depend on spatial correlation but on the total number of fractures. The probability for a fracture to belong to A_1 depends on spatial correlation; according to equation (3), it is proportional to $((l + l')/L)^D$.

[18] The probability of intersection within the area A_1, for a fracture of length l to connect the fracture of length l', is, however, not uniform. In a rough approximation, we can consider that connectivity occurs if the center of the smallest fracture is at a distance smaller than $l/2$ from the largest one (Figure 5). This defines an area $A_2 = l * l' + \pi(l/2)^2$ for a random orientation distribution, and a probability of intersection proportional to A_2/A_1. The exact calculation of A_2 can be done by following the method developed by *Balberg et al.* [1984].

[19] Eventually, the intersection probability $p(l, l' > l, L)$ is given by the product of two terms: the probability for a fracture to belong to A_1, and the probability of intersection within the area A_1 (equal to A_2/A_1). The general expression of $p(l, l' > l, L)$ is thus

$$p(l, l' > l, L) \propto \left(\frac{l' + l}{L}\right)^D \frac{A_2}{A_1} = \frac{(l' + l)^{D-2}}{L^D} \left(l^* l' + \pi\frac{l^2}{4}\right).$$

Note that the proportionality term of the above equation could be exactly derived for any distribution of orientation or of fracture shape [see, e.g., *Piggott*, 1997; *Huseby et al.*, 1997]. Its expected order of magnitude is ~ 1 in the case of large fracture shape eccentricity [*de Dreuzy et al.*, 2000]. This expression is in fact similar to classical excluded volume concepts with both a density term (the first one in the previous equation), and the excluded volume per fracture.

[20] If l is supposed to be much smaller than l', the expression becomes

$$p(l, l' > l, L) \propto \frac{l'^{D-1} l}{L^D}. \tag{5}$$

In three dimensions, the expression of A_2 and A_1 are $\pi(l'^2/4)l + \pi l'(l^2/4)$, and $4/3\pi[(l' + l)/2]^3$, respectively, leading to the general expression

$$p(l, l' > l, L) \propto \frac{(l' + l)^{D-3}}{L^D}\left(l'^2 l + l' l^2\right).$$

The expression is not symmetric with respect to l and l', and thus this prediction is valid only if $l' > l$. If $l = l'$, it gives $p(l, L) \sim (l/L)^D$ as expected from equation (4).

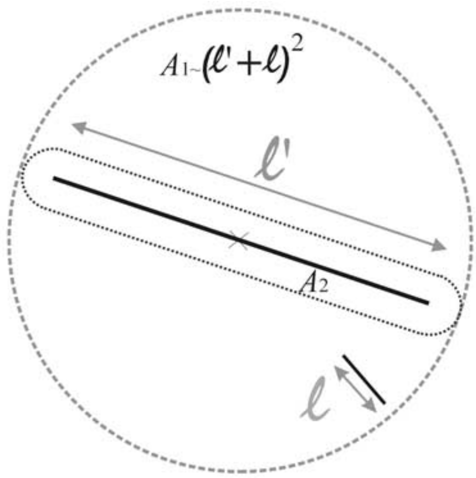


Figure 5. Representation of the areas A_1 and A_2 (dotted lines) used to determine the probability of intersection between fractures belonging to a fractal fracture network (equation (5)).

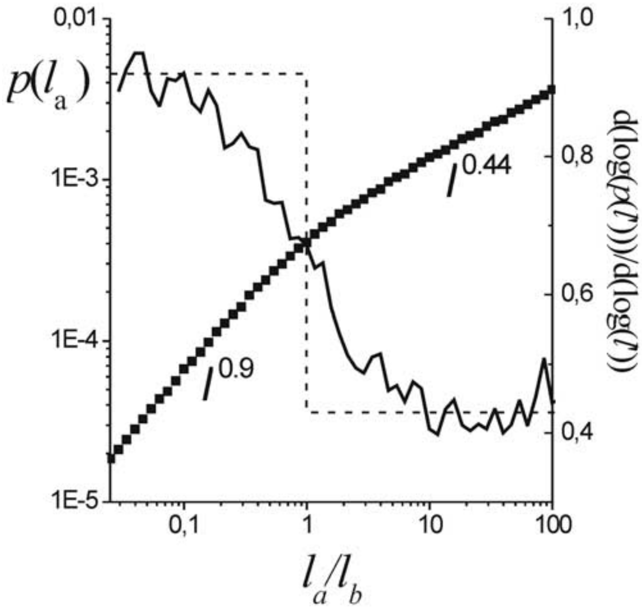


Figure 6. Computations of the intersection probability $p(l_a, l_b, L)$ between two fractures of length l_a and l_b for D = 1.3. For the numerical computations, l_b has been arbitrarily fixed to 40 while l_a varies in the range [1, 10000]. Monte Carlo simulations were performed in a system of size L = 10,000, for networks having 1000 fractures of size l_a and 1000 fractures of size l_b; $p(l_a, l_b, L)$ is plotted as a function of the ratio l_a/l_b (solid squares, left axis). The local variations of the slope of $\log[p(l_a, l_b, L)]$ versus $\log(l_a/l_b)$ (e.g., $d[\log p(l_a, l_b, L)]/d[\log (l_a/l_b)]$) is also represented (thin irregular curve, right axis) to better present the two regimes. The dashed line represents the theoretical model of equation (5), based on the two end-member limits for $l_a \ll l_b$, and $l_a \gg l_b$.

[21] Numerical computations of $p(l,l',L)$ have been performed in two dimensions for D between 1.3 and 2 and for systems of linear size $L = 10^4$. For different simulations, we generate two populations of fractures of respective length l_1 and l_2. The length l_1 has always been kept constant with $l_1 \ll L$, while the length l_2 varies within the range $[l_{min}; L]$. Numerically, the probability of intersection, $p(l_1, l_2, L)$ is deduced from the average number of intersections, $n_i(l_1, l_2, L)$, that a fracture of length l_1 has with all fractures of length l_2. To avoid finite size effects, we compute the number of intersections by unit length so that portions of fractures that lie outside the system are not taken into account in the computations.

[22] According to equation (5), the variation of $p(l_1, l_2, L)$ with l_2 is expected to change whether the ratio l_2/l_1 is smaller or larger than one. A typical example of one result is illustrated on Figure 6 for which D = 1.3. The two expected regimes are clearly identified for $(l_2/l_1) \ll 1$ and for $(l_2/l_1) \gg 1$. However, the transition between the two regimes spans two orders of magnitude (Figure 6). To highlight the transition, we also plot the local logarithmic slope with l_2 $(d(\log(p(l_1, l_2, L)))/d(\log(l_2)))$, which gives the apparent scaling of p with respect to l_2 (Figure 6, right-hand scale). Despite the broad transition, we obtain a result close to $p(l_1, l_2, L) \sim l_2$ for $l_2 < 0.1l_1$, and close to $p(l_1, l_2, L) \sim l_2^{D-1}$ for $l_2 > 10l_1$ in good agreement with (5).

[23] Similar results were obtained whatever the fractal dimension D (Figure 7). The agreement between equation (5) and calculations is fairly good (Figure 7), even if the theoretical prediction slightly overestimates or underestimates the scaling exponent if $l_2 \ll l_1$ or $l_2 \gg l_1$. We have not yet any explanation for this slight discrepancy, which increases when D decreases. We did check that the scaling exponents are really measured in the asymptotic regimes, and that this result is independent of the prefixed fracture length l_1, and of the number of fractures used in the simulations.

4.3. Consequences on the Connectivity of Fractures of Different Lengths

[24] In this section, we use previous results to derive the expression of the average number of intersections per fracture for a broad distribution of fracture lengths. We assume a length distribution such as equation (1) because it is representative of natural systems [*Odling*, 1997; *Bour et al.*, 2002]. The average number of intersections per fracture of length l is given by:

$$n_i(l, L) = \int_{l_{\min}}^{l_{\max}} p(l, l', L) \cdot n(l', L) dl',$$

where l_{min} and l_{max} are the smallest and largest fracture lengths in the system. According to the previous result, the

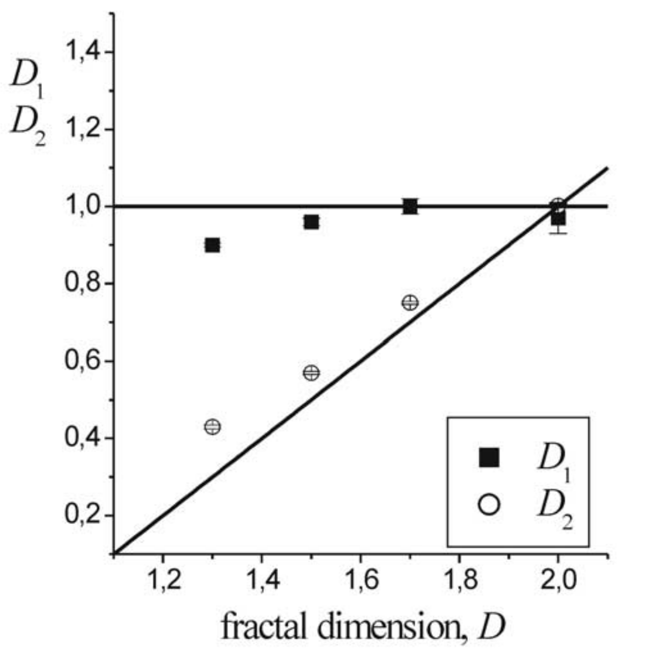


Figure 7. Numerical validation of expression (5) that relates the probability of intersection between two fractures of distinct lengths l_1 and l_2 to the fractal dimension D of the fracture network. For $l_1 \ll l_2$, $p(l_1, l_2, L)$ is expected to vary as $p(l_1, l_2, L) \sim l_1^{D_1} l_2^{D_2}$, with $D_1 \sim 1$ and $D_2 \sim D$-1. The corresponding exponents D_1 and D_2, derived from the numerical computations of $p(l_1 \ll l_2, l_2, L)$ (see Figure 6 for D = 1.3), are represented as a function of the fractal dimension D.

expression $p(l, l', L)$ changes whether l is larger, or smaller than l'. To derive analytical expressions for n_i, we use equation (5) as a simplified model of the intersection probability between fractures. We call it "the step model" to recall that it predicts a sharp crossover at $l = l'$.

[25] With this hypothesis, n_i becomes for two-dimensional fracture systems:

$$n_i(l,L) \approx \int_{l_{\min}}^{l} \frac{l^{D-1} l'}{L^D} n(l',L)dl' + \int_{l}^{L} \frac{l'^{D-1} l}{L^D} n(l',L)dl' + \int_{L}^{l_{\max}} \frac{l^{D-1} l}{L^D} n(l',L)dl'.$$

Combining the previous equation with (1) leads to

$$n_i(l,L) = \alpha\left(l^{D-1}\left[\frac{l'^{-a+2}}{(-a+2)}\right]_{l_{\min}}^{l} + l\left[\frac{l'^{-a+D}}{(-a+D)}\right]_{l}^{L} + lL^{D-1}\left[\frac{l'^{-a+1}}{(-a+1)}\right]_{L}^{l_{\max}}\right). \tag{6}$$

Note that a proportionality term should be added to take account of the distribution of orientation and of fracture shape.

[26] Depending on the value of the length distribution exponent, a, equation (6) may be simplified to the dominant terms so that:

$$\begin{aligned} n_{\mathrm{i}}(l,L) &\sim l^{D-1} && a>2,\\ n_{\mathrm{i}}(l,L) &\sim l^{D-a+1} && D<a<2,\\ n_{\mathrm{i}}(l,L) &\sim l && a<D. \end{aligned} \tag{7}$$

Equation (7) is an asymptotic regime of equation (6) that results from the dominance of one bound of the three integrals basic to this equation. It is valid only if the two bounds of the dominant integral are sufficiently different ($l \gg l_{\min}$ for $a > 2$; $l \ll L$ for $a < D$). In that case, we make the following comments from equation (7):

[27] 1. When fractures are uniformly distributed in space ($D = 2$), the number of intersections is simply proportional to their lengths in agreement with previous developments [*Bour and Davy*, 1997].

[28] 2. For a fractal fracture network, the behavior with l of $n_i(l, L)$ depends on the relative values of the exponents a and D. For $a > 2$, equation (7) predicts that the evolution with l of $n_{\mathrm{i}}(l, L)$ depends only on the fractal dimension D while it depends both on a and D for $D < a < 2$.

[29] 3. When the length distribution exponent a is too small, e.g., lower than D, the number of intersections is simply proportional to the fracture length, like in the case of uniformly distributed fracture networks ($D = 2$). Note that for a fractal network and for $a > D$, equation (7) implies that small fractures have more intersections per unit length than larger ones.

[30] 4. Equation (7) also suggests that it should be possible to derive the fractal dimension from the variations with l of $n_{\mathrm{i}}(l, L)$. In the discussion, we propose such an application on field data. Note that similar results are obtained when taking into account in the calculations fractures whose centers are located outside the system (see Appendix A).

[31] The same reasoning can be applied in three dimensions which leads to

$$\begin{aligned} n_{\mathrm{i}}(l,L) &\sim l^{D-1} && a>2,\\ n_{\mathrm{i}}(l,L) &\sim l^{D-a+1} && (D-1)<a<2,\\ n_{\mathrm{i}}(l,L) &\sim l^{2} && a<D-1, \end{aligned}$$

where $n_{\mathrm{i}}(l, L)$ is the number of intersections on a 2-D fracture and a and D are the length exponent and the fractal dimension, respectively, of the 3-D fracture network. As for 2-D systems, small fractures are proportionally better connected than larger ones in fractal fracture networks as long as the length exponent a remains larger than D-1.

[32] To check the preceding developments, we have computed the variation of the average number of intersections per fracture, $n_{\mathrm{i}}(l, L)$ for 2-D fractal fracture networks, for various values of D and a. For $a > 2$, equation (6) is dominated by the first term, and the scaling regime predicted by equation (7) is expected to occur if $l \gg l_{\min}$. Such behavior is indeed observed (Figure 8a), with scaling exponents that are reasonably well predicted by the asymptotic equation (7), even if slightly underestimated. Note that these slight discrepancies are consistent with those measured from $p(l, l', L)$ (Figure 7). For $a < D$, a linear variation of $n_{\mathrm{i}}(l, L)$ with l is expected and observed (Figure 8b). As predicted, this scaling regime, which comes from the second term of equation (6), is valid in the limit of small lengths compared to the system size ($l \ll L$). For $D < a < 2$, we expect $n_{\mathrm{i}}(l, L) \sim l^{D-a+1}$ from (7). In some cases, like for Figure 8b, we do observe such a scaling. However, in most cases we have an intermediate regime for which the exponents are not well defined, and seem to vary between D-1 and 1 when a varies from 2 to D. For these intermediate regime there is no clear dominant term in (6), as long as a is close to 2 or close to D.

[33] Despite slight discrepancies, we found that equation (5) well predicts the intersection probability between fractures and the average number of intersections per fracture of length l. In the following section, we still use equation (5) for deriving stereological rules for fractal fracture networks.

5. Stereological Properties of Fracture Networks

5.1. Analytical Developments

[34] For uniformly distributed fracture networks, the 3-D fracture length distribution may be easily derived from plane section observations. Indeed, the apparent fracture population of a d parent network sampled on a $(d-1)$-dimensional space (2-D plane, or 1-D transect) is simply given by the product $n_d(l, L)\,.\,p(l, L)$, where $p(l, L)$ is the probability of intersection between a fracture in the d-dimensional space and the sample of dimension $(d-1)$. For a uniform distribution of fracture densities (i.e., $D = d$), and for a power law distribution of disk diameters in 3-D, $p(l, L)$ is simply proportional to the fracture length l, so that the length distribution of the fractures over the plane is also a power law. The relationship between the exponents is summarized through [*Marrett and Allmendinger*, 1991; *Piggott*, 1997; *Berkowitz and Adler*, 1998]

$$a_d = a_{d-1} + 1. \tag{8}$$

The exponent of the 3-D power law length distribution $a_{3\text{-D}}$ is simply equal to $a_{2\text{-D}} + 1$, with $a_{2\text{-D}}$ the power law

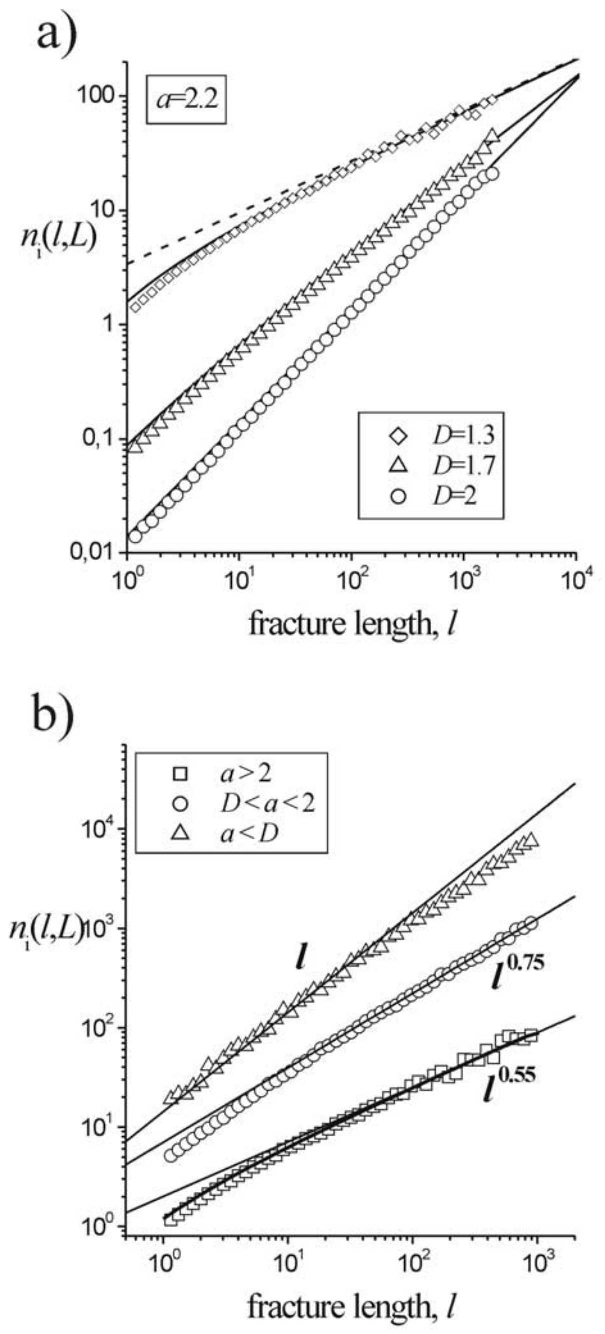


Figure 8. Scaling of $n_i(l, L)$, the average number of intersections per fracture, as a function of their length l, for several couples of parameters (a, D) (a) $a = 2.2$, $D = 1.3$, 1.7, and 2. The solid lines represent the theoretical expression given by equation (6). The values used for the exponents are the ones found in Figure 7 that are slightly different than 1 and D-1. For instance, the dashed line corresponds to $n_i(l, L) \sim l^{0.45}$ while the asymptotic behavior expected from equation (7) should be $n_i(l, L) \sim l^{0.3}$. (b) $D = 1.45$ and $a = 1.4$, 1.7, and 2.4. Straight lines correspond to $n_i(l, L) \sim l$, $n_i(l, L) \sim l^{0.75}$, and $n_i(l, L) \sim l^{0.55}$ for $a = 1.4$, 1.7, and 2.4, respectively. This scaling of $n_i(l, L)$ may be compared to the theoretical one given by equation (7).

exponent measured on any plane section. For a uniform distribution of fracture densities of the parent fracture network, D is simply equal to d so that we expect the distribution of fracture densities to be also uniform on lower dimensional sample, with $D_{3\text{-D}} = 3 = D_{2\text{-D}} + 1 = D_{1\text{-D}} + 2$.

[35] For fractal fracture networks (i.e., $D < d$), some changes in the stereological rules are expected since the probability of intersection between two elements is a function of the fractal dimension (equation 5). In the following, we aim at characterizing the relationship for both exponents a and D measured on the parent and on the intersected fracture network. We first characterize the length distribution of the intersecting fracture network.

[36] The length distribution that intersects a 1-D system such as a scan line of size L or a 2-D plane of size L^2 is given by

$$n_{d-1}(l,L) = p_d(l,l' = L,L) \cdot n_d(l,L),$$

where $n_d(l, L)$ and $n_{d\text{-}1}(l, L)$ are respectively the parent and the intersecting fracture length distributions. $p_d(l, l' = L, L)$ is the probability that a fracture of length l, belonging to a fracture network of fractal dimension D, intersects a sample of dimension $d-1$ and of size L. Assuming that a scan line crossing a 2-D fracture network can be assimilated to a large fracture of length L, $p_d(l, l' = L, L)$ is given by (5). Thus combining equations (1) and (5) for 2-D parent fracture networks leads to

$$n_{1-\mathrm{D}}(l,L) \approx \alpha L^{D_{2-\mathrm{D}}-1} l^{-a_{2-\mathrm{D}}+1} \qquad l < L,$$

$$n_{1-\mathrm{D}}(l,L) \approx \alpha L^{D_{2-\mathrm{D}}} l^{-a_{2-\mathrm{D}}} \qquad l > L.$$

The second equation expresses that a fracture larger than the system size always intersects the scan line ($p_d(l, l' = L, L) = 1$). Similarly the 2-D fracture length distribution resulting from the intersection of a 3-D network and a plane section is

$$n_{2-\mathrm{D}}(l,L) \approx \alpha L^{D_{3-\mathrm{D}}-1} l^{-a_{3-\mathrm{D}}+1} + \alpha L^{D_{3-\mathrm{D}}-2} l^{-a_{3-\mathrm{D}}+2} \qquad l < L,$$

$$n_{2-\mathrm{D}}(l,L) \approx \alpha L^{D_{3-\mathrm{D}}} l^{-a_{3-\mathrm{D}}} \qquad l > L.$$

[37] According to the previous developments, equation (8) should hold whatever the fractal dimension D for fractures smaller than the system size. On the opposite, for fractures larger than the system size, the length-distribution exponent of the intersecting fracture network should be the same as that of the parent distribution. Note that an additional complexity comes from fractures whose center lies outside the observation window, and which are not taken into account in the preceding developments. Their role is of course negligible for fractures having a length smaller than L with the same values of the length-distribution exponents. However, for fractures larger than L, it leads to a different expression (see Appendix A).

[38] The fractal dimension of the fracture centers cannot be simply inferred from the previous equations. It depends on the spatial distribution of all fractures whatever their length. The derivation of D can be accomplished by

calculating analytically the number of fractures intersecting the sample of size L:

$$N(L) \approx \int_{l_{\min}}^{l_{\max}} n(l, L) dl.$$

The scaling with L of $N(L)$ provides indeed the correlation dimension D (see below) [*Bour et al.*, 2002]. Note that the derivation of D through $N(L)$ can be done easily only if the fractures having a length larger than the system size can be neglected in the computations of the fractal dimension.

[39] For 1-D systems (scan line), the analytical development of $N(L)$ leads to:

$$N_{1-\mathrm{D}}(L) \approx L^{D_{2-\mathrm{D}}-1} \qquad a_{2-\mathrm{D}} \geq 2,$$

$$N_{1-\mathrm{D}}(L) \approx L^{D_{2-\mathrm{D}}-a_{2-\mathrm{D}}+1} \qquad D_{2-\mathrm{D}} \leq a_{2-\mathrm{D}}.$$

We expect therefore the fractal dimension along the scan-line to be equal to $D_{2\text{-D}} - 1$ or to $D_{2\text{-D}} - a_{2\text{-D}} + 1$, depending on the relative value of both $D_{2\text{-D}}$ and $a_{2\text{-D}}$. For $a < D$, the fractal dimension $D_{d\text{-}1}$ cannot be larger than the Euclidean dimension, here 1, which leads to the following stereological rules:

$$\begin{aligned} &D_{1-\mathrm{D}} = D_{2-\mathrm{D}} - 1 && a_{2-\mathrm{D}} \geq 2, \\ &D_{1-\mathrm{D}} = D_{2-\mathrm{D}} - a_{2-\mathrm{D}} + 1 && D_{2-\mathrm{D}} \leq a_{2-\mathrm{D}} \leq 2, \mathrm{k} \\ &D_{1-\mathrm{D}} = 1 && a_{2-\mathrm{D}} \leq D_{2-\mathrm{D}}. \end{aligned} \tag{9}$$

Equation (9) is consistent with the derivation of $n_i(l, L)$ whatever a and D (equation (7)). Note that analytical developments would have predicted an unrealistic scaling $N_{1\text{-D}}(L) \sim L^{-a+D+1}$ for $a < D$, with $D_{1\text{-D}} > 1$. In this case which is never encountered in natural systems, $N_{1\text{-D}}$ is dominated by fractures larger than the system size, and the scaling of $N_{1\text{-D}}$ conveys boundary effects due to limited system size rather than underlying fracture correlation. We overcome this problem in numerical computations by computing directly the fractal dimension of this set of points from a two-point correlation method.

[40] For 3-D systems, we derive the following relationships between $D_{2\text{-D}}$ and $D_{3\text{-D}}$:

$$\begin{aligned} &D_{2-\mathrm{D}} = D_{3-\mathrm{D}} - 1 && a_{3-\mathrm{D}} \geq 2, \\ &D_{2-\mathrm{D}} = D_{3-\mathrm{D}} - a_{3-\mathrm{D}} + 1 && D_{3-\mathrm{D}} - 1 \leq a_{3-\mathrm{D}} \leq 2, \\ &D_{2-\mathrm{D}} = 2 && a_{3-\mathrm{D}} \leq D_{3-\mathrm{D}} - 1. \end{aligned} \tag{10}$$

To conclude, the fractal dimension of a 1-D or a 2-D fracture network is not necessarily equal to $D_{d+1} - 1$ but can have any value in the range $[D_{d+1} - 1, d]$, depending on the length exponent a. On the other hand, equation (8) holds whatever the fractal dimension D, and it should be possible to simply derive the length distribution exponent.

5.2. Numerical Validation

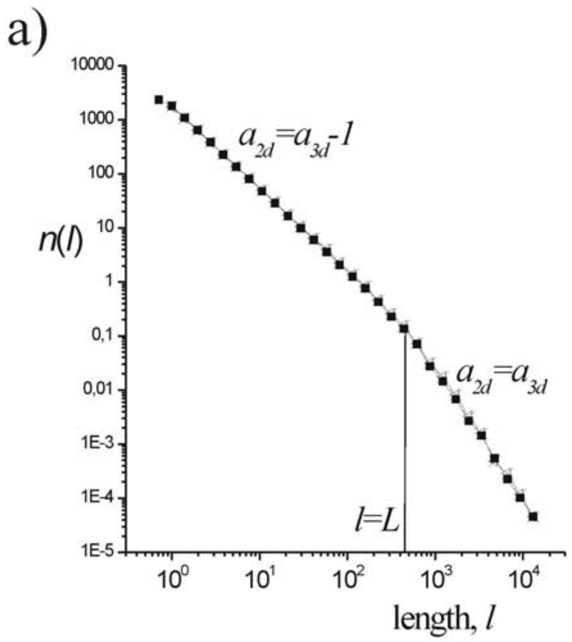
[41] To verify the preceding developments, numerical simulations have been performed from (1) a parent population of disc-shaped fractures (3-D network) and the corresponding population of 2-D fracture traces on a planar outcrop, and (2) a parent population of 2-D fracture lines and its 1-D intersection with a straight scan line (Figure 2). In both cases the linear size of the sample is equal to the system size L of the parent fracture network. The spatial fractal correlation is generated over 3 orders of magnitude, from a minimal scale equal to the length of the smallest fracture $l_{\min}$, fixed to 1 in practice, up to the system size L equal to 1024 and 512 for systems generated in two and three dimensions, respectively. The number of fractures of the parent network is increased until a sufficient number intersects the 1-D or 2-D sample. Then, the statistical geometrical properties of intersecting fracture networks are calculated. In practice, the statistical parameters are averaged over at least 100 simulations whatever the set of parameters (D, a) investigated.

[42] When the sample is a 1-D scan line, the sampled fracture network is theoretically made of the sole set of points that results from the intersection between the parent fracture network and the scan line (see example in Figure 1). The correlation dimension is derived from this set of points. The 1-D fracture length exponent can be derived only if the lengths of fractures that intersect the 1-D scan line are also registered. For well bore data, this information is generally not available. (Note, however, that *Borgos et al.* [2000] propose that the relationship between the fracture length and displacement could be used to infer fracture length from displacement measurement.)

[43] A typical example of the fracture length distributions stemming from the 2-D sampling of 3-D fractal fracture networks is given in Figure 9a for a length exponent $a_{3\text{-D}}$ equal to 2.5. The sampled fracture length distribution follows a power law whose scaling exponent depends whether the fracture lengths l are smaller or larger than the system size L. As expected from the theoretical analysis, the length exponent $a_{2\text{-D}}$ is equal to $a_{3\text{-D}}$ when $l > L$ since the probability to intersect the sample is about 1. However, for the relevant part of the analysis, e.g., when $l < L$, the length exponent $a_{2\text{-D}}$ is equal to $a_{3\text{-D}} - 1$ (Figure 9b). The numerical simulations show that equation (8), that predicts $a_{d\text{-}1} = a_d - 1$, still holds for any value of the fractal dimension D (Figure 9b). This result means simply that the intersection probability is proportional to fracture length whatever the fractal dimension of system. This result may not be valid if there is a correlation between fracture length and position as suggested by *Ackerman and Schlische* [1997] and *Bour and Davy* [1999].

[44] To examine the expected dependency of the fractal dimension with the length exponent a (see equations (9) and (10)), we computed the correlation dimension of the intersecting fracture networks from the pair correlation function $C(r)$, defined as the number of pairs of points, amongst all possible pair combinations, whose distance is smaller or equal to r [*Viseck*, 1992]. In a fractal system, $C(r)$ varies like r^D, where D is the fractal dimension. $C(r)$ is measured from the distribution of the centers of length traces in two dimensions, and from the set of intersecting points along the scan-line in one dimension.

[45] The method is illustrated in Figure 10a for different fractal dimensions of parent fracture networks. The correlation function $C(r/L)$ and the variation with r of the local slope ($d[\log(C(r/L))]/d[\log(r/L)]$, right-hand scale) are simultaneously represented. All the curves are normalized



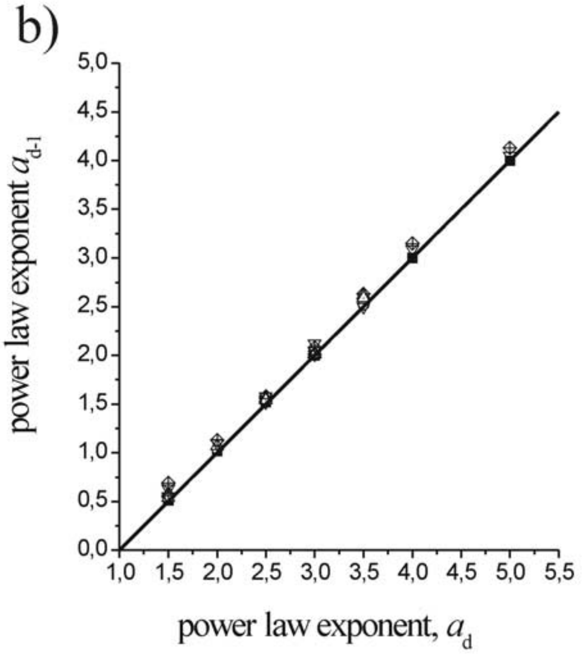


Figure 9. (a) Example of a length distribution of a 2-D trace fracture network resulting from the intersection of a 3-D fracture network with a 2-D plane. For $l < L$, the power law length exponent $a_{2\text{-D}}$ is equal to $a_{3\text{-D}}$-1, whereas $a_{2\text{-D}} = a_{3\text{-D}}$ for $l > L$, $a_{3\text{-D}}$ being the exponent of the power law length distribution of the parent fracture network. (b) Exponents $a_{d\text{-}1}$ measured from the density length distribution (for $l < L$, see Figure 9a). The computations are made for the transition from three to two dimensions and for the transition from two to one dimension. We have $a_d = a_{d\text{-}1} + 1$ whatever the fractal dimension D_d of the parent fracture network.

such that $C(1) = 1$ for $r = L$ according to *Bour et al.* [2002]. Over the example shown, all curves $C(r/L)$ can be fitted with power laws over almost two orders of magnitude. Finite size effects are observed only if r/L is larger than 0.5.

[46] In Figures 10b and 10c the derived fractal dimensions $D_{d\text{-}1}$ are represented as a function of the parent fractal dimension D_d. The theoretical relation $D_{d-1} = D_d - 1$, expected for $a_\text{d} > 2$ (equations (9) and (10)) is verified. The value of $D_{d\text{-}1}$ is, however, slightly overestimated in a way that is consistent with the deviations observed in the connectivity between fractures (Figure 7). This discrepancy with theoretical predictions is less pronounced for $D_{2\text{-D}}$ versus $D_{3\text{-D}}$ (Figure 10b) than for $D_{1\text{-D}}$ versus $D_{2\text{-D}}$ (Figure 10c). In the worst case, e.g., for a close to 2 and D close to 1, the differences are about 0.2.

[47] For $a_{3\text{-D}} < D_{3\text{-D}} - 1$ (Figure 10b) or for $a_{2\text{-D}} < D_{2\text{-D}}$ (Figure 10c), we observe $D_{d-1} = d-1$, in agreement with previous analytical developments. In that cases, the number of intersections due to fractures having a length close or larger than the system size is very large leading to an apparent uniform density distribution of the fracture network.

[48] In the intermediate regime, corresponding to $D_{3\text{-D}} - 1 < a_{3\text{-D}} < 2$ in 3-D, and to $D_{2\text{-D}} < a_{2\text{-D}} < 2$ in two dimensions, we observe some variations of the derived fractal dimension that are approximately in agreement with previous analytical developments. The derived values of $D_{d\text{-}1}$ are ranging between D_d -1 and d -1; however, the precise determination of $D_{d\text{-}1}$ is rather difficult in that intermediate range of a and D values, so that we do not obtain exactly the expected behavior $D_{d\text{-}1} = D_d - a_d + 1$ (equations (9) and (10)).

[49] To sum up, we have characterized the stereological rules in two or three dimensions that apply to the fracture model defined by equation (1). Results derive from the expression (5) that predicts the probability of intersection of a small fracture with a larger one, so that they can be extrapolated beyond the model (1).

5.3. Discussion: Application to Field Data

[50] The previous stereological analysis has shown that a 3-D fracture length distribution can be theoretically derived from plane section observations: a power law distribution observed from plane section fracture traces should result from a power law distribution for 3-D fracture lengths, even for stochastic fractal fracture networks. The extrapolation in three dimensions is obtained by assuming that the probability to observe a fault trace in the plane section is proportional to the trace length l whatever the fractal dimension of the fracture network, as it is the case for uniform fracture networks. This is equivalent to assume that the trace length l is the characteristic length scale of the fracture, even for 3-D correlated fracture networks. The exponent of the 3-D power law length distribution $a_{3\text{-D}}$ is thus expected to be equal to $a_{2\text{-D}} + 1$, with $a_{2\text{-D}}$ the power law exponent measured on any plane section.

[51] In contrast, *Malinverno* [1997] and *Borgos et al.* [2000] found different stereology equations for the power law length exponents, $a_{2\text{-D}} = a_{3\text{-D}} - D + 2$ according to our notations, with a model which has also fractal correlation. In their object model, however, the fractal correlations are basically defined from the distance r between object centers

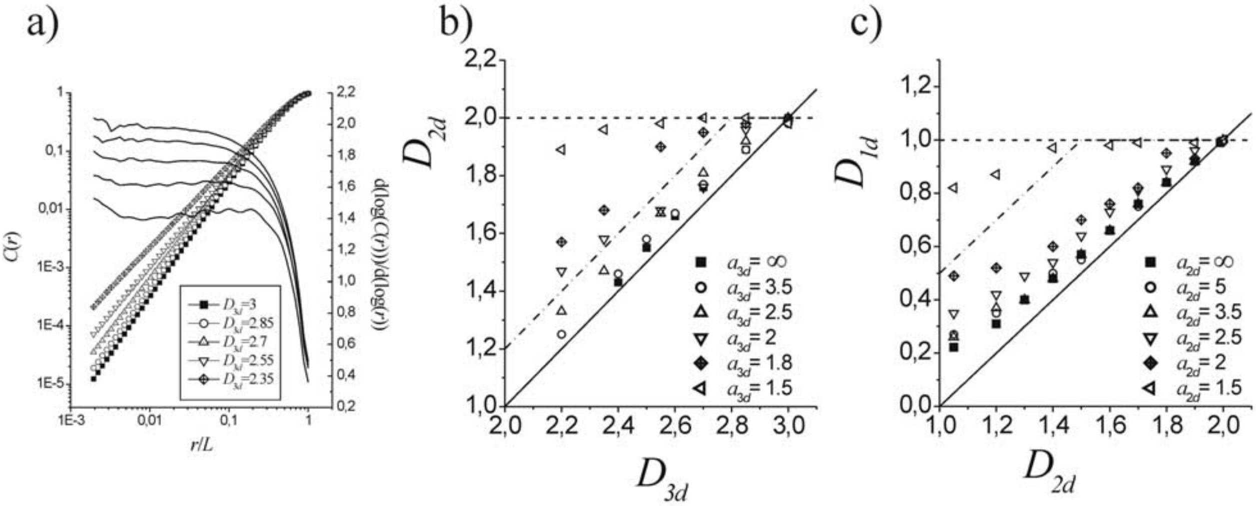


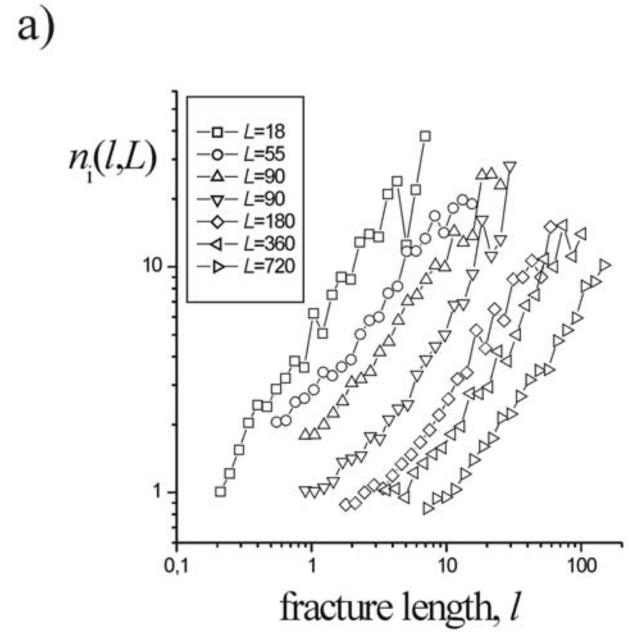
Figure 10. Derivation of the fractal dimension of fracture networks resulting from the 2-D sampling of a 3-D parent network (Figures 10a and 10b), and from the 1-D sampling of a 2-D parent network (Figure 10c). (a) Correlation function $C(r/L)$. The value of $D_{2\text{-D}}$ may be visually estimated from the solid lines which give the local value of the slope (right-hand scale) of the log $C(r)$ − log r plot. (b) Derived value of $D_{2\text{-D}}$ as a function of $D_{3\text{-D}}$. (c) Derived value of $D_{1\text{-D}}$ as a function of $D_{2\text{-D}}$. In Figures 10b and 10c, thick solid lines correspond to $D_{d-1} = D_d - 1$, while dashed horizontal line corresponds to $D_{\text{d}-1} = d - 1$. In both case, the dash-dotted line corresponds to the theoretical value of $D_{d\text{-}1}$ (see equations (9) and (10)) for $a_d = 1.5$ (Figure 10b) and $a_d = 1.8$ (Figure 10c).

and the sampling scan line, such that the average object density measured in the plane perpendicular to the scan line decreases as r^{D-2}. Their model is thus self-affine rather than self-similar since the scaling along the scan line is different from the scaling in both other perpendicular directions. Also the scan line is not independent of the object structure. In contrast our fractal model is isotropic, and the scan line is randomly chosen with respect to the general fracture network so that there is no expected fracture density decrease with the distance to the scan line. Both these differences are sufficient to explain the differences in terms of stereological rules. Orientation distributions are also different but this does not change the stereological relationships between exponents.

[52] From our results, we expect $a_{3\text{-D}}$ to vary between 2 and 5 for natural fault networks [*Segall and Pollard*, 1983a, 1983b; *Gudmunsson*, 1987; *Villemin and Sunwoo*, 1987; *Childs et al.*, 1990; *Main et al.*, 1990; *Scholz and Cowie*, 1990; *Gautier and Lake*, 1993; *Davy*, 1993; *Odling*, 1997]. Since $a_{3\text{-D}}$ is expected to be larger than 2 in most cases, we expect $D_{3\text{-D}}$ to be given through $D_{3\text{-D}} = D_{2\text{-D}} + 1$. Note that since we found slight discrepancies between $D_{2\text{-D}}$ and $D_{3\text{-D}} - 1$ (Figure 10b) $D_{3\text{-D}}$ may be slightly lower than $D_{2\text{-D}} + 1$. Such a derivation cannot be done easily from one-dimensional data sets since there is in general no information on fracture lengths. The extrapolation of measured fractal dimensions to 2- or 3-D systems is therefore speculative knowing that $a_{2\text{-D}}$ has been often measured close or lower to 2 [*Bonnet et al.*, 2001]. However, in the absence of reliable 3-D data sets, a comparison between 1- and 2-D data sets can be nevertheless very useful to test the applicability of our results to natural systems. To our knowledge, there has never been such a direct comparison, the scaling laws being established either from 1- or 2-D data sets [*Bonnet et al.*, 2001].

[53] Here, we use a data set from the Hornelen Basin (Norway) that consists of seven 2-D fracture patterns mapped in the same area from the meter scale up to almost the kilometer scale [*Odling*, 1997]. Its main geometrical properties can be described at all scales by the simple model described in equation (1) with $a_{2\text{-D}} = 2.75 \pm 0.05$, $D_{2\text{-D}} = 1.8 \pm 0.04$ and $\alpha = 3.5$ [*Bour et al.*, 2002]. From the above stereological rules, we expect the fractal dimension $D_{1\text{-D}}$ along a scan line to be slightly larger than 0.8. To analyze the 1-D properties of the Hornelen fracture network, we applied the procedure suggested above from equation (7), that consists in computing the average number of intersections per fracture as a function of their length, $n_{\text{i}}(l, L)$ and deriving its scaling with l. Since the measurements of $n_{\text{i}}(l, L)$ is an average over the number of fractures of a given length, this procedure should provide a reliable derivation of $D_{1\text{-D}}$, that should not be too sensitive to the undersampling of small fractures. For each fracture map, we indeed obtain a data set that may be approximated by a power law (Figure 11a). A power law fit over the whole range of lengths provides exponents that vary from 0.75 to 0.93, depending on the fracture data set. In most cases, the exponent derived is close to 0.84. Despite some variability in the measurements, all derived values are slightly larger than 0.8 (except one) and smaller than 1, as expected.

[54] A proper normalization of the different curves should permit the comparison between maps in order to estimate the validity of a single power law model for the whole data set. Because of sampling and resolution effects, the minimal cutoff fracture length, $l_{\min}$, observable on each map is dependent on the system size [*Odling*, 1997]. Assuming that $l_{\min}$ varies proportionally to the system size L, equation (6) can be used to derive the variations with L of the average number of intersections per fracture so that $n_{\text{i}}(l, L) \sim l^{D-1}L^{-a+2}$. Therefore the computed curves



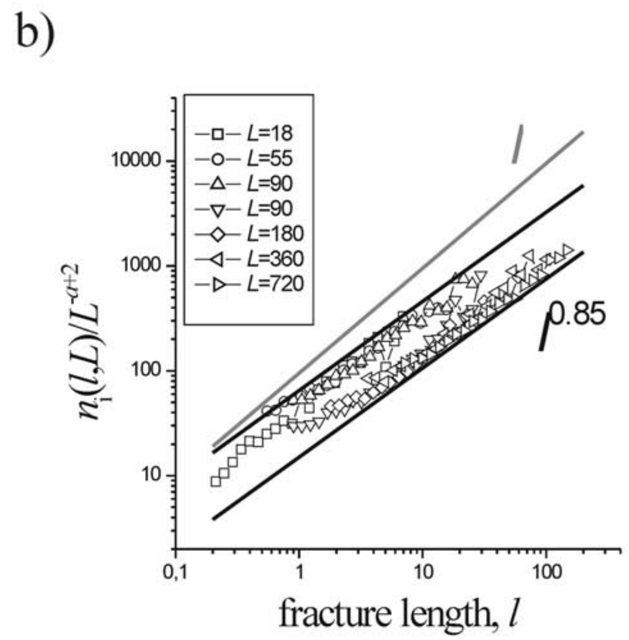


Figure 11. (a) Evolution of the average number of intersections per fracture, $n_i(l, L)$, as a function of their length l, for the different fracture maps of the Hornelen Basin fracture network [*Odling*, 1997; *Bour et al.*, 2002]. (b) Same curves as in Figure 11a normalized by L^{-a+2} (see text).

$n_i(l, L)/L^{-a+2}$ should be properly normalized for the different fracture maps. The normalized curves are indeed approximately well lined up (Figure 11b). From the normalized curves, it appears clearly that the variation of $n_i(l, L)$ with l does not match the function $n_i(l, L) \sim l$ corresponding to $D_{1\text{-D}} = 1$. Although it remains difficult to estimate precisely $D_{1\text{-D}}$, the best power law model provides an estimate of $D_{1\text{-D}}$ slightly larger than 0.8 (Figure 11b). This result, obtained thanks to a new method for deriving the fractal dimension, appears therefore consistent with the previous analytical and numerical results.

[55] In complement, we should point out that N. E. Odling (unpublished data, 1999–2001) obtained $D_{1\text{-D}} \sim 0.86$ on three large 1-D data sets also coming from the Hornelen Basin. The three data sets were obtained by measuring fracture intersections along three long scan lines. Using also the two-point correlation function, *Du Bernard et al.* [2002] obtained 1-D values of $D_{1\text{-D}}$ ranging between 0.85 and 0.91 for five 1-D data sets sampled in the Suez rift.

6. Conclusion

[56] To assess the stereological rules of fractal fracture networks, we first analyzed the connectivity properties between fractures. We show how the intersection probability of fractures of different lengths, $p(l, l', L)$, depends on the fractal dimension D. From the analytical expression of $p(l, l', L)$, we derive the expression of the stereological rules that extrapolate the parameters of the 3-D length distribution from their measurements on 2-D fracture networks. We assume a power law length distribution of fractures but our results can easily be extended to other length distributions. Our analytical results were checked through a numerical model that permits us to generate 3-D discrete fractal fracture networks.

[57] We show that, for the length distribution exponent a, the relation $a_{d-1} = a_d - 1$ is valid whatever the fractal dimension D. In contrast, the extrapolation of the fractal dimension depends on the value of the length exponent a. For $a_{2\text{-D}}$ or $a_{3\text{-D}}$ larger than 2, we have a simple relation: $D_{d-1} = D_d - 1$. For $a_{2\text{-D}} < D_{2\text{-D}}$ or $a_{3\text{-D}} < D_{3\text{-D}} - 1$, we have $D_{d-1} = d - 1$, meaning that the fracture density distribution in the plane sample or along the scan line is uniform. This peculiar behavior is due to the greater number of fractures having a size close or larger than the system size when the exponent a decreases toward 1. For such large fractures, the location of the intersection in the 2-D plane or along the scan line is no more related to the position of the parent fracture center. As a consequence, the intersecting fracture network (fracture traces in two dimensions, and set of points in one dimension) is no more clustered (when $a_{2\text{-D}} < D_{2\text{-D}}$ or $a_{3\text{-D}} < D_{3\text{-D}} - 1$), although the parent fracture network is fractal. For $D_{2\text{-D}} < a_{2\text{-D}} < 2$ or $D_{3\text{-D}} - 1 < a_{3\text{-D}} < 2$, we observe an intermediate regime for which $D_{d\text{-}1}$ varies between $D_d - 1$ and $d - 1$.

[58] Our results show that in practice the derivation of the length exponent is required for extrapolating the fractal dimension in two or three dimensions. It therefore severely limits the interest of one-dimensional data sets, such as well bore data, for which only fracture densities distribution is available. Nevertheless, a fractal dimension measured in one dimension with $D_{1\text{-D}} < 1$, implies the fracture network to be fractal in two dimensions or in three dimensions, even if the extrapolation of $D_{2\text{-D}}$ and $D_{3\text{-D}}$ is speculative.

[59] We finally discuss the applicability of our results to natural systems. We found a good adequacy between our predictions and measurements made on few natural data sets. In addition, we propose also an original method for measuring the fractal dimension of natural data sets from the

variations of the average number of fracture intersections per fracture of length l.

Appendix A: Role of Large Fractures Generated Outside the System

[60] To determine the influence of large fractures that have their center outside the observation window, and that nevertheless crosscut the 2-D outcrop or the 1-D scan line, we first derive the fracture length distribution, $n_{obs}(l, L)$, that is observed in a subdomain of linear size L. We assume that all fractures belong to a very large 2-D domain of size Λ and we check for the length distribution, $n_{obs}(l, L)$, in a 2-D subdomain of size L; $n_{obs}(l, L)$ is given by

$$n_{obs}(l, L) = n(l, \Lambda) p_{\cap}(l, L, \Lambda),$$

where $n(l, \Lambda)$ is the power law length distribution of the fracture network in the system of size Λ and $p_{\cap}(l, L, \Lambda)$ is the probability of intersection between a fracture of length l and the subdomain of observation of linear size L.

[61] For fractures $l \ll L$, we assume that a fracture is observed within the subdomain only if its center lies into the subdomain. From the definition of a fractal object [*Mandelbrot*, 1982] that provides the scaling of the number of boxes of size L necessary to cover the fractal (see section 4.1), it follows $p_{\cap}(l < L, L, \Lambda) \sim (L/\Lambda)^D$. For fractures $l \gg L$, a fracture whose center is outside the subdomain has nevertheless a nonnegligible probability to intersect the subdomain. Assuming that the probability that a subdomain of size L intersects a large fracture of size $l \gg L$ is equivalent to the probability of intersection between two fractures of size l and L, $p_{\cap}(l < L)$ should be given by (5), so that $p_{\cap}(l > L) \sim (Ll^{D-1})/\Lambda^{D}$. Both expressions of $p_{\cap}$ for $L \gg l$ and $L \ll l$ should therefore lead to

$$n_{obs}(l, L) \propto L l^{D_{2\text{-D}} - a_{2\text{-D}}} \qquad l \ll L$$

$$n_{obs}(l, L) \propto L l^{-a_{2\text{-D}} + D_{2\text{-D}} - 1} \qquad l \gg L.$$

Using these equations for deriving the average number of intersections per fracture, $n_i(l, L)$, we obtain the same scaling than in equation (7). For the stereological problem, the length distribution that intersects a 1-D scan line of size L is obviously modified for $l > L$, for which we have $n_{1\text{-D}}(l, L) = \alpha L l^{-a+D-1}$. However, taking into account this modification leads to equation (9) for the derivation of the fractal dimension through the computation of the total number of fractures along the scan line.

[62] Similar calculations may be done in three dimensions. The length distribution, $n_{obs}(l, L)$, of a 3-D subdomain of size L, is thus given by

$$n_{obs}(l, L) \propto L l^{-a_{3\text{-D}} + D_{3\text{-D}} - 1} \qquad l \ll L$$

$$n_{obs}(l, L) \propto L l^{D_{3\text{-D}} - a_{3\text{-D}} - 1} + L^2 l^{D_{3\text{-D}} - a_{3\text{-D}} - 2} \qquad l \gg L.$$

[63] Using now the probability of intersections between a fracture of length l and a fracture of length l' in a system of volume L^3, $p(l, l', L)$ and knowing that $p(l, L, L) \sim 1$ for $l > L$, it provides the length distribution of fractures that intersect the 2-D sample. The scaling of $N_{2\text{-D}}(L)$ with L does not provide the fractal dimension $D_{2\text{-D}}$ for $a_{3\text{-D}} < D_{3\text{-D}}$, due to a too large proportion of long fractures having a length larger than the system size. For $a_{3\text{-D}} < D_{3\text{-D}}$ and especially for $a_{3\text{-D}} < 2$, most fractures on the 2-D sample have their center outside the system. The intersecting fracture networks are therefore equivalent to fracture networks made of fractures of infinite length for which $N_{2\text{-D}}(L)$ varies linearly with L.

Notation

l	fracture length.
L	size of the fracture network domain.
$n(l, L)$	density length distribution.
$N(L)$	number of fractures included in a system of linear size L.
d	Euclidean dimension.
D	fractal dimension.
a	exponent of the density length distribution.
$p(l, l', L)$	probability of intersection between two fractures of characteristic length l and l' in a system of characteristic linear size L.
$n_i(l, L)$	averaged number of intersections per fracture of length l.

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